

The Two-Time Moving-Defect Problem for Dispersing Billiards

Dominant-frequency correlation, physical regeneration, and a pathwise CM2 theorem

Working note

12 July 2026 (version 26)

Abstract

We isolate the two-time estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of dispersing scatterers. The moving first defect contains regular two-dimensional terms and singular one-dimensional physical faces. Neither a projective-cone trace estimate nor a nuclear graph-current representation is available. We avoid both. In the central two-time band we first remove contracted coordinates and then truncate only the two scalar dominant frequencies of each assembled physical face. The retained finite sum is a separately dynamically Hölder function of finitely many states of the billiard process. The functional-correlation theorem for finite-horizon dispersing billiards therefore gives exponential decay with a polynomial dominant-frequency loss. The complementary dominant modes are treated by the physical phase estimate; no ambient Fourier vector is incorrectly identified with a tangent frequency. Off the central band a signed proper-family theorem supplies forward and reverse coupling. This removes graph Banach spaces, tensor renewal, and nuclear decompositions of the regular defect.

The high-frequency face term is controlled by a physical slope energy. We use Leclerc's native unstable coordinate and his cutting theorem modulo every fixed-degree polynomial. A physical regeneration theorem uses the growth lemma and clean-magnet minorization to feed actual collision-SRB standard families into legal native children; it is not inferred from a smooth extension of a fractal horseshoe conditional. A single-source conditional identity freezes one oriented endpoint and places the other endpoint in the context polynomial. A bounded-overlap native stopping cover, a mass-preserving resolving buffer, and Jensen give the reduced-slope small-ball estimate. The choice $\sigma = r^{1/(1+\alpha_L)}$ separates Leclerc's cutting exponent from the billiard $C^{1+\alpha_B}$ remainder. On the nonresonant set a direct $C^{1+\alpha}$ fractional nonstationary-phase estimate yields a polynomial Fourier saving, which becomes exponential after a cutoff $K = e^{\eta \max\{m,n\}}$ in the balanced band.

For radial faces we index every connected component and normalize its mass before forming the boundary functional. A two-chart arclength calculation is combined with a weighted flux-complexity condition; no component is silently discarded. The resulting standard families recover after $O(\log Z)$ iterates. Qualitative residual transversality is insufficient for the infinite-depth limit, so the theorem is stated for analytic or circular table deformations satisfying an explicit pathwise jet-tail condition with a strict return/grazing margin. A depth $L(s) \asymp \log(1/|s|)$ is used: shallow faces remain transverse along the coarea segment and the deeper difference quotient is an $o(1)$ scalar tail. Under the primitive hyperbolicity, actual-SRB regeneration, flux-complexity, and pathwise transversality inputs listed below, the complete first defect satisfies CM2.

1 The exact target

Throughout, $\{T_s : |s| < s_0\}$ is a C^3 compact family of finite-horizon planar dispersing billiard collision maps generated by a C^3 normal deformation of strictly convex scatterer boundaries and transported to one reference collision space M . The chosen analytic-boundary or finite-dimensional circular deformation is subject to QT_{path} , specified in Section 7.3. The table at $s = 0$ contains the unobstructed clean cycle used in Proposition 6.8. We consider only the complete *first* moving-table defect: regular Jacobian/density derivatives and all same-label physical one-sided faces are assembled before centering. The tests f, g belong to $C^r(M)$, with r above the Fourier summability threshold, and $\mu_0(g) = 0$. Second shape defects and unassembled point atoms are outside the statement.

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. For the difference quotient it is enough to have the same estimate on words of depth at most $L(s)$ together with an $o(1)$ CM2 tail beyond $L(s)$; this is the pathwise diagonal criterion below.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \mapsto w(h|_{I^+} - h|_{I^-}) \delta_\Gamma, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C\varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C\varepsilon|W|^{-q} \leq C\varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C\varepsilon|y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C\chi^m \implies \|K_V P_0^m(f_+ - f_-)\|_Y \leq C'\chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *Used globally, a one-gap functional-correlation estimate gives only $O(\tau^m + \vartheta^n)$ and does not by itself prove CM2. It is nevertheless exactly sufficient in the central band, where $m \asymp n$: after smoothing the event kernel, one decoupling gives $e^{-c \max\{m, n\}}$, hence an $e^{-c'(m+n)}$ bound there. Forward and reverse standard-family coupling handle the two off-diagonal regions; Lemma 6.2 assembles the product estimate. Thus no operator-ideal estimate is used in the principal proof.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (6) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j(W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C 2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

4.1 Recovery windows and the actual summability target

For differentiability of the susceptibility, a uniform pointwise CM2 bound for every grazing block is stronger than necessary. The required conclusion is absolute summability of the complete block series. This distinction removes an artificial exponential loss in the grazing rank.

Proposition 4.1 (Recovery-window summability). *Let $B_b(m, n)$ be the scalar pairing produced by block b at past and future times m, n . Suppose that there are masses M_b , recovery times R_b^-, R_b^+ , constants $C < \infty$, and $0 < \tau, \vartheta < 1$ such that*

$$|B_b(m, n)| \leq CM_b \tau^{(m-R_b^-)_+} \vartheta^{(n-R_b^+)_+}, \quad m, n \geq 0. \quad (11)$$

Then

$$\sum_{b, m, n} |B_b(m, n)| \leq C' \sum_b M_b (1 + R_b^-)(1 + R_b^+). \quad (12)$$

In particular, if $M_b \leq C2^{-\alpha b}$ for some $\alpha > 0$ and $R_b^\pm \leq C_0 + C_1 b$, then the full threefold series converges. The same conclusion holds uniformly for parameter difference quotients if the constants and block masses are uniform.

The deterministic recovery times may be replaced by a countable refinement $B_b = \sum_j B_{b,j}$ with masses $M_{b,j}$ and times $R_{b,j}^\pm$, provided

$$\sum_j M_{b,j} (1 + R_{b,j}^-)(1 + R_{b,j}^+) \leq CM_b (1 + b)^2. \quad (13)$$

It is enough for the two-sided recovery tail, after its deterministic $O(b)$ entrance time, to be exponential.

Proof. For every $R \geq 0$ and $0 < q < 1$,

$$\sum_{k \geq 0} q^{(k-R)_+} \leq R + 1 + \frac{1}{1-q}.$$

Sum (11) first in m, n and then in b . Exponential block mass dominates the quadratic recovery window. Applying the same calculation to every refined piece gives (13). An exponential tail for $R_{b,j}^- + R_{b,j}^+ - O(b)$ has uniformly bounded polynomial moments, proving the last assertion. $\square$

Lemma 4.2 (Two marginal growth tails imply bi-recovery). *Let a block of total variation M_b be cut by the common refinement of its past and future recovery trees, with refined masses $M_{b,j}$ and recovery times $R_{b,j}^\pm$. Assume cutting preserves total variation and, for some $r_b \leq C_0 + C_1 b$, $0 < \rho < 1$,*

$$\sum_{j: R_{b,j}^\pm > r_b + t} M_{b,j} \leq CM_b \rho^t, \quad t \geq 0,$$

for each sign. Then (13) holds.

Proof. If $R_{b,j}^- + R_{b,j}^+ > 2r_b + t$, at least one marginal time is larger than $r_b + t/2$. The union bound therefore gives an exponential tail for the sum of the two recovery times under the normalized weights $M_{b,j}/M_b$. Its second moment is $O((1+r_b)^2)$. Since $(1+R^-)(1+R^+) \leq (1+R^-+R^+)^2$, the claim follows. $\square$

For a general graph current the relevant geometric input is therefore a *proper bi-recovery* statement, not an estimate of a Fourier constant on the original grazing curve. For the clean first radial current this input is Corollary 6.44. A central two-time theorem then gives (11). The fully unrecovered rectangle is handled by total variation, the two mixed rectangles by the corresponding recovered one-sided estimate, and the fully recovered rectangle by the product bound. The growth-lemma tail sums these windows through (13).

Consequently even the generic $M_b = O(2^{-b})$ assembly margin is sufficient once proper bi-recovery is proved; the stronger first-radial $O(2^{-(3-\epsilon_*)b})$ margin gives additional room but is no longer needed to absorb an exponential Fourier constant in b . This decouples the grazing rank from the return-rank truncation in Proposition 6.4.

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (14)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (15)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (15) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (16)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (16), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (16) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (16) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b \gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m} \gamma_b(t)) f(T^n F_b \gamma_b(t)) w_b(t) dt. \quad (17)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b \gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (18)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_* \nu_b}(\xi)| \leq C_b (1 + |\xi|)^{-\eta} \quad (19)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (20)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(19) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

This preliminary proposition is not used as an assumption in the final proof. Uniform (19) is supplied below by the cohomological local slope-energy theorem, the dominant-coordinate reduction, and the physical fractional nonstationary-phase estimate. One-sided mixing alone would not supply it.

6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole (m, n) quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

Lemma 6.2 (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (21)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}, \quad (22)$$

with all four exponents positive. Choose $A > \beta_-/\alpha_+$ and $D > \beta_+/\alpha_-$. Suppose in addition that

$$|B_{m,n}| \leq C e^{-\gamma \max\{m, n\}}$$

whenever $D^{-1}m < n < Am$. Then there are $\tau, \vartheta \in (0, 1)$ such that $|B_{m,n}| \leq C' \tau^m \vartheta^n$ for all m, n .

Proof. If $n \geq Am$, (21) is bounded by $e^{-c_+(m+n)}$ for some $c_+ > 0$. If $m \geq Dn$, use (22). The remaining indices lie in the stated diagonal band, where $\max\{m, n\} \geq (m+n)/2$. Taking the minimum of the three positive exponents proves the claim. $\square$

For billiards, (21) follows from signed standard-family coupling after cutting the past-weighted density into homogeneous pieces: the growth lemma gives the coupling factor $e^{-\alpha+n}$, while differentiating the m -step past weight costs at most $e^{\beta-m}$. The marked-amplitude moment sums the homogeneous pieces. Applying the same argument to the reversible collision map gives (22). All constants are uniform on the fixed compact atlas; recovered grazing pieces enter only after their recorded recovery times; these are the signed standard-family forms of the usual growth and coupling estimates [20, 2]. Consequently the genuinely new slope/Fourier theorem is needed only for $m \asymp n$.

This also corrects the naive ambient phase test. The graph $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$ is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable–unstable temporal-distance/holonomy function. A finite-difference slope-value QNL is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.3 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (23)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (23) follows. On the tangency face, $|(d, 0) - q_j| \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in the reduced phase

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m}\gamma(t) + \zeta \cdot T^n F\gamma(t) \quad (24)$$

have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.4 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq Ce^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq Ce^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus this fallback would require a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails could then remove the truncation, but the truncation-uniform constants are not supplied by that route. The direct gate-resolved slope argument below bypasses this fallback entirely.

6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an η_{cut} fraction of its parent's conditional mass. For an arbitrary exhaustion by finite subshifts, η_{cut} may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank L . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (25)$$

with η_0 independent of x, L , for at least one QNL-separated child at each scale. If (25) and a uniform UNI constant $\kappa_0 > 0$ hold, Leclerc’s tree argument gives a QNL exponent $\Gamma > 0$ independent of L ; only the multiplicative constant is allowed to grow like e^{cL} .

Rühr’s pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (25) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

Lemma 6.5 (Uniform interval UNI implies power non-concentration). *Let ν_L be measures on an interval whose densities satisfy $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$, uniformly in L . Let X_L be a family of functions with the following scale-invariant UNI property: for every interval J of length $\sigma < \sigma_0$ and every polynomial P in a fixed finite-dimensional class, there is a subinterval $J' \subset J$ with $|J'| \geq \kappa|J|$ such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where $\kappa > 0$ is independent of J, P, L . Then for every fixed $\alpha > 0$ there are $C, \Gamma > 0$, independent of L , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

Proof. Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval J' . The density bounds imply that at least a fixed fraction $\eta_0 > 0$ of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to $|J|^{1+\alpha}$. The number of generations is $c_\alpha |\log |J|| + O(1)$, so the surviving mass is at most $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C|J|^\Gamma \nu_L(J)$. $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

6.5 A fixed two-branch UNI certificate

Bertozi’s 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let h_0, h_1 be two regular inverse return branches on the central magnet, with a common contraction bound $\lambda < 1$, and let τ denote the reduced stable–unstable phase increment (for the geometric application, $\tau = \log J^u F$ after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (26)$$

Proposition 6.6 (Anchored two-branch certificate). *Suppose the series in (26) converges in C^1 and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

on a central unstable interval I . Then, for all sufficiently large N ,

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.

Proof. C^1 convergence makes $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$ for all large N . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation. $\square$

Corollary 6.7 (First-term domination after acceleration). *Assume $\|Dh_i\| \leq \lambda < 1$, $\|D\tau\|_\infty \leq M$, and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

If

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

then $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$.

Proof. The derivative of the sum of all terms with $k \geq 2$ is bounded by $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1-\lambda)$. Subtract this from the first-term lower bound. $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (23); attach fixed regular connectors to the magnet; and compute Ψ'_∞ (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter Ψ_∞ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order λ for the first-term curvature separation; the remaining infinite temporal-distance tail is only $O(\lambda^2)$. This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not C^2 . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates, write the map as (F, G) with derivative $\text{diag}(\lambda, \mu)$, where $\lambda > 1$, $\mu = \lambda^{-1}$. The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (27)$$

with all derivatives evaluated at the fixed point.

Proposition 6.8 (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let $\mathcal{R} = T^2$ be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates (s, p) with $p = \sin \varphi$. Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write $S^{-1}\mathcal{R}S = (F, G)$. Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} &= F_{yy} = G_{xx} = G_{xy} = G_{yy} = 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

Proof. Place the circle centres at $(0, 0)$ and $(3, 0)$ and parametrize the first boundary near the facing point by $q(s) = (\cos s, \sin s)$. For outgoing Birkhoff data (s, p) , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at q and a target circle with centre c and radius one, put $w = q - c$, $b = w \cdot v$, and $c_0 = |w|^2 - 1$. The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates (s, p) to $(-s, -p)$ and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by S gives $G_{xyy} = -(7 - 4\sqrt{3})$; substitution in (27) gives -1 . $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength s/R and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and $\mathfrak{A}$ depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let W^u be a reference unstable leaf and let $H : \Gamma \rightarrow W^u$ be stable holonomy from a central smooth face arc. Write m_Γ for arclength and let w include the physical face velocity, coarea factor, and the smooth branch Jacobians.

Lemma 6.9 (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i) H is a $C^{1+\alpha}$ diffeomorphism onto its image and $0 < c_H \leq JH \leq C_H$;
- (ii) the SRB conditional μ^u on W^u has arclength density between $c_u > 0$ and $C_u < \infty$;
- (iii) $w \in C^\alpha(\Gamma)$ and $\|w\|_\infty \leq C_w$.

Define the signed face measure

$$\nu = H_*(w dm_\Gamma).$$

If, on every interval $J \subset W^u$, the temporal-distance phase X satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing C_1 by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with C_1 multiplied by the sum of their total-variation bounds.

Proof. The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for μ^u compare arclength and μ^u in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces. $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. For pieces whose return word or homogeneity rank tends to infinity, this local transfer is combined below with the central marked moment and the two-sided recovery sum; Proposition 7.6 keeps those constants on one common refinement.

6.8 A countable-IFS theorem removes the return truncation

There is a more direct route than passing through smooth finite horseshoes. Leclerc–Paukkonen–Sahlsten [17] prove power Fourier decay for Gibbs measures of finite or countable $C^{1+\alpha}$ conformal IFSs, allowing overlaps. In the countable case their assumptions are uniform contraction, bounded distortion, summable variations, and the light-tail condition

$$\sum_a \sup_{x \in I} e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} < \infty \quad \text{for some } \gamma_0 > 0. \quad (28)$$

Their Theorem 1.1 says that QNL gives power Fourier decay, including for Hölder amplitudes and nonstationary phases with a small polynomial loss in their norms. Sections 6.1–6.2 prove QNL by finite-difference cutting arguments for finite MNL and for countable UNI systems with a light tail. No finite-alphabet exhaustion or smooth extension of the branch maps is used.

This theorem is unusually well matched to the present induced problem. Let $F : I \rightarrow I$ be the unstable quotient of a full-branch Gibbs–Markov return to a central billiard magnet and let $\{h_a : I \rightarrow I\}_{a \in \mathcal{A}}$ be its inverse branches. After accelerating once if necessary, the desired dictionary is

countable-IFS input	billiard input
h_a	homogeneous inverse return branch
ϕ	normalized induced unstable-Jacobian potential
uniform contraction	uniform hyperbolicity after inducing
bounded distortion	homogeneous return distortion
light tail	return tail plus a small grazing moment
QNL	a fixed clean finite-difference gate.

(29)

There is a small but important point in the first line of this dictionary. The one-sided moving-face labels must not be inserted by cutting every return branch into labelled subbranches: those pieces need not have full image and would destroy the hypothesis to which the countable-IFS theorem is applied. The correct object is a full-branch dynamical quotient with a marked amplitude fiber.

For an interval amplitude use the affine scale-normalized norm

$$\|w\|_{\mathcal{A}^\alpha(J)} := \|w\|_{\infty, J} + |J|^\alpha [w]_{C^\alpha(J)}. \quad (30)$$

It is unchanged up to a fixed factor by affine normalization and cannot increase under restriction. Every connected component is charged separately; hence endpoint and component complexity is visible in the sum rather than hidden in a piecewise-Hölder notation.

Proposition 6.10 (Marked full-branch quotient). *Let Λ be a Young return rectangle with product structure and let $\{\Lambda_a\}_{a \in \mathcal{A}}$ be a return partition such that $F|_{\Lambda_a} = T^{R_a}$ maps the stable subset Λ_a across Λ with the usual stable/unstable Markov property. Fix a reference unstable leaf I and let $\pi^s : \Lambda \rightarrow I$ be stable holonomy. Then*

$$\bar{F} \circ \pi^s = \pi^s \circ F$$

defines a one-state full-branch expanding map: for $I_a = \pi^s(\Lambda_a)$, the map $\bar{F} : I_a \rightarrow I$ is onto and has an inverse branch $h_a : I \rightarrow I_a$.

Suppose now that the lift of a complete physical face through Λ_a is refined into one-sided marked pieces $E_{a,\ell}$, after artificial-face cancellation. On the stable quotient its action can be written

$$\nu(\varphi) = \sum_{a,\ell} \int_{J_{a,\ell}} \varphi(h_a x) w_{a,\ell}(x) d\mu^u(x), \quad (31)$$

where the $J_{a,\ell} \subset I$ are amplitude pieces and the sign is included in $w_{a,\ell}$. Thus the quotient dynamical alphabet remains $\{h_a\}_{a \in \mathcal{A}}$; the face label ℓ is a fiber mark and does not have to define a full-image branch. If

$$\sum_{a,\ell} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty,$$

then the marked current is a summable piecewise-Hölder amplitude over the same full-branch Gibbs system.

Proof. The Markov property says that every unstable leaf in Λ_a is mapped across every stable leaf of Λ . Quotienting stable leaves therefore makes $\bar{F}|_{I_a}$ onto I ; contraction of inverse branches and bounded distortion are the usual quotient estimates. Disintegrate the coarea face measure first over return atoms and then over its one-sided physical marks. Stable holonomy transports each piece to the reference leaf and gives (31). No restriction $h_a|_{J_{a,\ell}}$ is declared to be a dynamical branch. The last assertion is the triangle inequality for the Hölder-amplitude norm. Matched artificial cuts contribute opposite weights and disappear before this disintegration. $\square$

Consequently the standard Young quotient and the moving-face bookkeeping are compatible provided the marked-amplitude sum is finite. The construction of the quotient is standard. Proposition 6.11 supplies this summability on the central return alphabet, while the bi-recovery refinement supplies the grazing tail.

The SRB potential part of this audit is now supported by a direct billiard result. Lemmas 5.2–5.3 of Lima–Obata–Poletti [19] prove that the collision Lebesgue/SRB measure is an equilibrium state of the geometric potential

$$\phi = -\log \|DT|_{E^u}\|, \quad P(\phi) = 0,$$

and that ϕ becomes Hölder on their countable Markov coding for finite-horizon dispersing billiards. Hence the induced potential has summable variations. This does not by itself give the one-state return tail, but it removes the former ambiguity between an MME coding and the required collision-SRB Gibbs weights.

Proposition 6.11 (Central amplitude moment). *Assume the Young return scheme has*

$$\sum_a \sup_I p_a e^{\epsilon R_a} < \infty$$

and restrict first to a compact regular homogeneity core. For a complete first moving-face defect, suppose every collision insertion has uniformly bounded C^α coarea and stable-holonomy factors. Then the marked amplitude disintegration (31) satisfies

$$\sum_{a,\ell} e^{\epsilon' R_a} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty \quad (0 < \epsilon' < \epsilon).$$

After high-homogeneity insertions are assigned to the bi-recovery blocks of Corollary 6.44, the same conclusion holds for the entire assembled first defect.

Proof. A return word of length R_a has at most R_a possible physical insertion times. On the compact regular core each insertion contributes at most a uniform constant times the Gibbs branch weight. Hence the left-hand side is bounded by

$$C \sum_a R_a e^{\epsilon' R_a} \sup_I p_a,$$

which is finite because $Re^{\epsilon' R} \leq C_{\epsilon, \epsilon'} e^{\epsilon R}$. Artificial insertions cancel before taking total variation. The removed high-homogeneity pieces satisfy the common two-sided recovery moment and are summed by Proposition 4.1. $\square$

The following consequence records the exact countable-IFS interface rather than hiding it in a finite-horseshoe approximation.

Proposition 6.12 (Finite-difference countable-IFS bridge). *Let the central unstable quotient be a one-state full-branch $C^{1+\alpha}$ IFS with uniform inverse contraction, bounded distortion, a normalized Gibbs potential of summable variations, and (28). For words $\mathbf{a}, \mathbf{b}$ of length N and $x, y \in I$, put*

$$\Delta_N((\mathbf{a}, x), (\mathbf{b}, y)) = \{\log |h'_{\mathbf{a}}(x)| - \log |h'_{\mathbf{b}}(x)|\} - \{\log |h'_{\mathbf{a}}(y)| - \log |h'_{\mathbf{b}}(y)|\}.$$

Assume the LPS quantitative nonlinearity estimate: for some $C > 0$, $\Theta > 0$, and $\rho \in (0, 1)$,

$$\sum_{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{A}^N} p_{\mathbf{a}} p_{\mathbf{b}} p_{\mathbf{c}} p_{\mathbf{d}} \mathbf{1}_J(\Delta_N((\mathbf{a}, x_{\mathbf{c}}), (\mathbf{b}, x_{\mathbf{d}}))) \leq C\{|J|^{\Theta} + \rho^N\} \quad (32)$$

for every interval $J \subset \mathbb{R}$; Gibbs weights may be evaluated at arbitrary reference points in their cylinders. Then the induced SRB Gibbs measure has the weighted nonstationary-phase Fourier estimate of Leclerc–Paukkonen–Sahlsten for every phase with the stated C^{α} log-derivative bounds. The conclusion remains valid for the marked physical face amplitudes of Lemma 6.9.

Proof. Equation (32) is precisely the QNL hypothesis in Theorem 1.1 of [17]. That theorem is formulated for finite or countable $C^{1+\alpha}$ systems and Hölder amplitudes; it neither differentiates $\log h'_a$ nor assumes a C^2 holonomy. The light-tail moment controls the countable alphabet. The weighted-face conclusion follows from Lemma 6.9 and the summable scale-normalized amplitude norm (30). $\square$

The former C^2 multiplier/UNI route is deliberately not used. For a billiard quotient the stable holonomy is generally only $C^{1+\alpha}$, so the derivative of $\log h'_a$ need not exist. The correct gate input is the finite-difference estimate (32).

Proposition 6.13 (Clean homoclinic horseshoe supplies a finite QNL gate). *Let p be the regular fixed point of the squared billiard map associated with the normal period-two orbit in Proposition 6.8. If its Anosov cocycle satisfies $\mathfrak{A}(p) \neq 0$, then there is a compact finite clean horseshoe Ω_p , contained in regular branches and containing p , such that in at least one time orientation its temporal distance satisfies QNL for every Hölder equilibrium state. Thus Ω_p supplies a finite topological QNL seed; Proposition 6.19 transfers its polynomial escape to the physical collision-SRB bracket.*

Proof. Lemma 5.1 of Lima–Obata–Poletti [19] says that finite-horizon dispersing billiards have a single homoclinic class at regular hyperbolic points. Choose a second regular hyperbolic periodic point q . The theorem gives clean transverse intersections $W^u(p) \pitchfork$

$W^s(q)$ and $W^u(q) \cap W^s(p)$. The inclination lemma then gives a nontrivial clean transverse homoclinic point of p ; this avoids the vacuous substitution $x = y = p$. The Smale–Birkhoff construction, using a finite segment of this clean homoclinic orbit, gives a compact finite horseshoe Ω_p on which the billiard map is smooth. In symplectic Birkhoff coordinates the map is area preserving; the local dynamics near Ω_p may be viewed as a smooth Axiom–A basic set.

By the contrapositive of the Anosov cocycle obstruction, together with Theorem 4.3 of Leclerc [16], in one of the two time orientations the temporal distance on Ω_p satisfies QNL for every Hölder equilibrium state, in particular for the equilibrium state of the geometric potential restricted to this finite subsystem. No comparison of that horseshoe measure with collision SRB is made here. $\square$

There is a regularity caveat. The stable quotient of a smooth horseshoe is in general only $C^{1+\alpha}$, precisely because the stable holonomy need not be C^2 . Therefore the finite QNL conclusion above cannot be fed directly into the C^2 non-linearizability criterion of Algom–Rodriguez Hertz–Wang. The route below instead propagates the physical gate’s finite-difference gap to the countable $C^{1+\alpha}$ inducing scheme. The LPS theorem is recorded as a background alternative; it is not invoked in the principal fractional-phase proof.

The light-tail hypothesis has a concrete billiard reduction. If a branch a has return time R_a and grazing rank k_a , bounded distortion gives, schematically,

$$|h'_a|^{-\gamma_0} \leq C e^{c\gamma_0 R_a} (1 + k_a)^{q\gamma_0}.$$

Thus an exponential return moment and any strictly positive polynomial SRB tail in k_a imply (28) after choosing $\gamma_0 > 0$ sufficiently small. This is strictly weaker than uniform control of every rank- L smooth extension.

Lemma 6.14 (Return and grazing moments imply the LPS light tail). *Assume the normalized branch weights $p_a(x) = e^{\phi(h_a x)}$ satisfy*

$$\sum_a \sup_x p_a(x) e^{\epsilon R_a} (1 + k_a)^\delta < \infty$$

for some $\epsilon, \delta > 0$, and suppose $|h'_a|^{-1} \leq C e^{cR_a} (1 + k_a)^q$. Then (28) holds for every

$$0 < \gamma_0 < \min\{\epsilon/c, \delta/q\}.$$

Proof. Raise the inverse-derivative estimate to γ_0 and absorb the two factors into the assumed exponential and polynomial moment. $\square$

There is also a pressure-based way to obtain the required moment. The geometric-potential theory for finite-horizon dispersing billiards has a spectral gap for $-t \log J^u T$ on every compact subinterval of $(0, t_*)$, where $t_* > 1$, and its modified one-step expansion is uniform for $t \geq t_0 > 0$ [2]. Since the SRB value is $t = 1$, the LPS factor $|h'_a|^{-\gamma_0}$ merely changes the geometric exponent from 1 to $1 - \gamma_0$.

Proposition 6.15 (Geometric-pressure route to the light tail). *Let a homogeneous full-branch inducing scheme have an exponential return tail at the SRB parameter. Assume its induced distortion is uniform and that the geometric transfer estimates hold for $t \in [1 - \epsilon, 1]$. Then, after decreasing $\epsilon > 0$ if necessary, its normalized induced SRB potential satisfies (28) for every sufficiently small $\gamma_0 > 0$.*

Proof. Group branches by return time $R_a = n$. Up to the uniform distortion coboundary, the summand in (28) is the inverse unstable Jacobian to the power $1 - \gamma_0$. The geometric transfer estimate at $t = 1 - p\gamma_0$ bounds the corresponding p -moment by $Ce^{c(p\gamma_0)n}$, with $c(p\gamma_0) \rightarrow 0$ as $\gamma_0 \rightarrow 0$. Hölder's inequality combines this with the SRB return tail Ce^{-an} to give

$$\sum_{R_a=n} \sup_x e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} \leq Ce^{-an/q + c(p\gamma_0)n/p}.$$

Choose $p, q > 1$ conjugate and then γ_0 so that the exponent is negative; summing over n proves the claim. Homogeneity-strip subdivision is already included in the geometric one-step expansion. $\square$

Lemma 6.16 (Fixed-gate hitting and two-context moments). *Let F be a one-state full-branch Gibbs–Markov map and let $\mathcal{G}$ be a fixed finite gate cylinder of return depth L . Suppose bounded distortion gives*

$$\inf_C \mu^u(\mathcal{G} \mid C) \geq \eta_G > 0 \quad (33)$$

for every past cylinder C , and suppose

$$\mathbb{E}e^{\epsilon R} < \infty \quad (34)$$

for some $\epsilon > 0$. If N_G is the first gate-hitting time in L -blocks and C^+ is the total collision length of the future entrance context, then there is $\epsilon_G > 0$ such that

$$\mu^u(N_G > k) \leq (1 - \eta_G)^k, \quad \mathbb{E}e^{\epsilon_G C^+} < \infty.$$

The reverse quotient gives the same estimate for C^- . On the natural extension no independence is needed:

$$\mathbb{E}e^{\epsilon'_G (C^- + C^+)} < \infty \quad (35)$$

for every sufficiently small $\epsilon'_G > 0$.

Proof. Condition (33) and the Markov property give $\mu^u(N_G > k + 1 \mid N_G > k) \leq 1 - \eta_G$, proving the geometric tail. Let $K(\theta)$ be the uniform conditional exponential moment of one return; $K(0) = 1$ and $K(\theta) < \infty$ for small $\theta > 0$. Bounded Gibbs distortion gives

$$\mathbb{E}\left[e^{\theta C^+}; N_G = k\right] \leq C\{(1 - \eta_G)K(\theta)^L\}^{k-1}K(\theta)^L.$$

Choose θ so that the quantity in braces is smaller than one and sum in k . Time reversal proves the past estimate. Finally

$$\mathbb{P}(C^- + C^+ > t) \leq \mathbb{P}(C^- > t/2) + \mathbb{P}(C^+ > t/2),$$

so the two marginal exponential tails imply (35) without independence. $\square$

For the marked quotient, choose one fixed finite gate word. The Gibbs ratio of that word after an arbitrary prefix differs from its unconditional weight by at most the uniform distortion factor. Hence (33) holds with $\eta_G = C_D^{-1}\mu^u(\mathcal{G}) > 0$; it is not an additional mixing assumption.

Lemma 6.17 (Joint return–maximal-grazing moment). *Let Y be the base of an exponential Young return, R its return time, and let $B(x)$ be the largest homogeneity rank visited before $R(x)$. In $(r, p) = (r, \sin \varphi)$ coordinates suppose*

$$\mu\{1 - |p| \lesssim 2^{-2b}\} \leq C2^{-2b}, \quad \mu_Y(R > L) \leq Ce^{-cL}.$$

Then B has an exponential tail. More precisely, for every fixed $A > 0$,

$$\mu_Y(B \geq b) \leq Ce^{-cAb} + CA b 2^{-2b}. \quad (36)$$

Consequently, for sufficiently small $\epsilon, \kappa > 0$,

$$\mathbb{E}_Y e^{\epsilon R + \kappa B} < \infty. \quad (37)$$

If the inverse-Jacobian and coarea loss of a return word is bounded by $Ce^{c_0 R + qB}$, then (28) and the marked-amplitude moment hold after decreasing the exponents. No independence of R and B is required.

Proof. On $\{R \leq Ab\}$ take a union bound over the at most Ab collision times. Since $\mu_Y(E) \leq \mu(E)/\mu(Y)$ for every event E and the collision SRB measure is invariant, each visit to a rank- b boundary layer costs at most $C2^{-2b}$. Adding $\mu_Y(R > Ab)$ proves (36). Choose A so that both terms decay exponentially in b ; hence $\mathbb{E}e^{q\kappa B} < \infty$ for some $q > 1$ and small κ . For conjugate $p, q > 1$, Hölder gives

$$\mathbb{E}e^{\epsilon R + \kappa B} \leq (\mathbb{E}e^{p\epsilon R})^{1/p} (\mathbb{E}e^{q\kappa B})^{1/q} < \infty.$$

Absorb $c_0 R + qB$ with smaller exponents and apply Lemma 6.14. The same argument applies to the scale-normalized coarea amplitude. $\square$

Lemma 6.17 verifies the grazing factor from the collision density and strip width, while Proposition 6.11 supplies the moving-face amplitude budget on the same quotient. Lemma 6.16 supplies the entrance/exit moment. Thus the particular full-branch quotient uses one joint return/grazing/context ledger rather than an unverified appeal to the literature theorem.

This route also exposes a flaw in the finite-horseshoe detour. A conservative pasting lemma [21] can locally modify or smooth a volume-preserving map, but it does not automatically extend a finite family of billiard branches to a global area-preserving Axiom–A diffeomorphism whose equilibrium state has the required collision-SRB weights. Even a successful extension can put a fractal Gibbs conditional on a saddle basic set in place of the absolutely continuous unstable conditional of the billiard SRB measure. The direct countable-IFS quotient preserves the correct measure and should therefore be the primary route.

Thus light-tail is not expected to be a new principal theorem once the standard homogeneous inducing scheme is put in the one-state form required above. The exact value $\mathfrak{A} = -1$, the clean homoclinic theorem, and Proposition 6.13 produce the clean polynomial escape seed without naming a connector. Proposition 6.24 records the local physical slope transfer, and the cohomological bracket below identifies its physical collision-SRB object.

6.9 The cohomological bracket and polynomial context blow-ups

The quotient regularity is only $C^{1+\alpha}$. The useful exact object is therefore a four-point difference of Birkhoff sums, not a derivative of a holonomy. Let f denote the billiard map

restricted to the finite clean basic set, let F be its stable factor, and put $\tau_f = \log J^u f$ and $\tau_F = \log |F'|$. As in [16, Lemma 3.15], choose a Hölder function θ such that

$$\tau_f = \tau_F \circ \pi + \theta \circ f - \theta. \quad (38)$$

For p, q in one Markov rectangle define

$$\Delta(p, q) = \sum_{j \in \mathbb{Z}} \{ \tau_f(f^j p) - \tau_f(f^j[p, q]) - \tau_f(f^j[q, p]) + \tau_f(f^j q) \}. \quad (39)$$

Lemma 6.18 (Cohomological bracket identification). *Let $\mathbf{a}, \mathbf{b}$ be the length- N past words of p, q and let $x = \pi p$, $y = \pi q$. With inverse branches $g_{\mathbf{a}}, g_{\mathbf{b}}$ of F , set*

$$\begin{aligned} \mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) &:= S_N \tau_F(g_{\mathbf{a}} x) - S_N \tau_F(g_{\mathbf{a}} y) \\ &\quad - S_N \tau_F(g_{\mathbf{b}} x) + S_N \tau_F(g_{\mathbf{b}} y). \end{aligned} \quad (40)$$

Then

$$|\mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) - \Delta(p, q)| \leq C \kappa^{\alpha N}. \quad (41)$$

The constants are uniform on the persistent finite basic set and for $|s| < s_0$. Entrance and exit contexts are not included in this statement; their finite jets and holonomy tails are treated explicitly in Proposition 6.19.

Proof. Insert (38) in (39). The four θ -terms telescope. For $j \geq 0$, stable projection gives $\pi(f^j p) = \pi(f^j[p, q])$ and $\pi(f^j q) = \pi(f^j[q, p])$, so the positive-time factor terms cancel exactly. Writing the negative iterates through the past inverse words gives (40). The omitted tail starts at $N + 1$; Hölder regularity of τ_F and inverse contraction give

$$\sum_{j > N} C \kappa^{\alpha j} \leq C' \kappa^{\alpha N}.$$

This is precisely the proof of [16, Lemma 3.15] on the clean basic set. Hyperbolic continuation makes the constants uniform in s . $\square$

The next statement transfers the all-scale escape in [16, Propositions 4.31–4.32] from the horseshoe conditional to the physical collision-SRB conditional. This is the only support transfer that is required.

The preceding bracket must still be matched to the physical endpoint maps. For a stable projection π_s on a clean rectangle, the stable holonomy Jacobian satisfies the convergent identity

$$\log J \pi_s(r) = \sum_{j \geq 0} \{ \tau_f(f^j r) - \tau_f(f^j \pi_s r) \}. \quad (42)$$

This is the Jacobian formula of [16, Lemma C.1]. In particular, entrance and exit holonomies are not discarded as unspecified coboundaries: their four endpoint series are inserted into (39); the common terms cancel and the unpaired tail is bounded by $C \kappa^{\alpha N}$.

Proposition 6.19 (Context-polynomial blow-up). *Assume $\mathfrak{A}(p) = -1$ and let $I_c = t_c + \ell_c[-1, 1]$ be any homogeneous resolving cylinder whose central excursion enters the persistent clean basic set. For the full reduced-slope bracket S_c , including the two endpoint holonomy Jacobians, let N be the total central resolving depth and let $L_c = |c^-| + |c^+|$ be the entrance/exit length. There are a fixed degree d , a rough template $\mathcal{X}_{\omega_c}$ from the clean basic set, a polynomial P_c of degree at most d , and a context cost $H_c \geq 1$ such that*

$$\ell_c^{-1} \{ S_c(t_c + \ell_c u) - S_c(t_c) \} = \mathcal{X}_{\omega_c}(u) + P_c(u) + E_c(u), \quad \|E_c\|_{\infty} \leq C \left(H_c \ell_c^{\alpha} + \frac{\kappa^{\alpha N}}{\ell_c} \right). \quad (43)$$

Moreover $H_c \leq Ce^{aL_c}$, and for some $p > 1$ the context ledger gives

$$\sup_{|s| < s_0} \sum_c w_c H_c^p < \infty. \quad (44)$$

The degree and the escape constants of $\mathcal{X}_{\omega_c}$ modulo $\mathbb{R}_d[X]$ are independent of c and s . In particular, if

$$N \geq q_* |\log \ell_c|, \quad q_* > \frac{1 + \alpha}{\alpha |\log \kappa|}, \quad (45)$$

then the second error in (43) is $O(\ell_c^\alpha)$.

Proof. Use (42) for both endpoint maps and split each series at the entrance and exit times. The four common tails cancel in the bracket. Each remaining finite clean context sum is $C^{1+\alpha}$ on its homogeneous component. Its affine Taylor polynomial is linear (and is regarded as an element of $\mathbb{R}_d[X]$); the normalized remainder is bounded by its $C^{1+\alpha}$ norm times ℓ_c^α . The chain rule and uniform one-step distortion give $H_c \leq Ce^{aL_c}$. Choose $p > 1$ so that pa is smaller than the exponent in (35); this proves (44). Lemma 6.18 identifies the central part with the factor four-point cocycle with error $O(\kappa^{\alpha N})$. Condition (45), rather than an unrecorded fixed acceleration, makes its normalized value $O(\ell_c^\alpha)$. Finally [16, Propositions 4.31–4.32], activated by $\mathfrak{A}(p) \neq 0$, gives uniform all-location, all-scale escape of the rough template from every polynomial of degree at most d . No fixed acceleration is used. $\square$

6.10 Native-coordinate single-source slope energy

The following two conditions isolate the genuinely new clean-gate geometry on the *actual* collision-SRB interval. They are not inferred from a smooth extension of a fractal horseshoe conditional.

Definition 6.20 (Single-source bracket condition SS). *Fix one orientation and condition on the opposite symbolic record, the value t' , and the two finite contexts. For a native interval Q write $z = \Phi_{\beta,Q}(t)$ and $\sigma_Q = \text{diam } \Phi_{\beta,Q}(Q)$. Put $d_Z = d$, where d is the fixed degree in Proposition 6.19. Condition SS(d_Z) means that*

$$S_\beta(t) - S_\beta(t') = \ell_{\beta,Q} \{X_{\beta,Q}(z) - P_{\beta,Q,t'}(z)\} + \mathcal{R}_{\beta,Q,t'}(z), \quad (46)$$

where $P_{\beta,Q,t'} \in \mathbb{R}_{d_Z}[z]$, $X_{\beta,Q}$ is one native Leclerc rough template, and

$$0 < c_\ell \leq |\ell_{\beta,Q}| \leq C_\ell, \quad \|\mathcal{R}_{\beta,Q,t'}\|_\infty \leq C \{H_\beta \sigma_Q^{1+\alpha_B} + \kappa^N\}. \quad (47)$$

The endpoint identity proving that every other t -dependent rough term is absent or belongs to the fixed-degree polynomial is part of the condition; freezing a symbolic name alone is not enough.

Definition 6.21 (Actual-SRB physical regeneration PR). *Let $\mu_{\beta,Q}^u$ be the normalized collision-SRB conditional on a native interval Q . Condition PR(α_L, d_Z) consists of the following three primitive billiard statements.*

(PR1) *In the native coordinate the scaled density of $\mu_{\beta,Q}^u$ is uniformly bounded above and below and has the standard scaled log-Hölder bound.*

- (PR2) *After at most $C \log(1/\sigma_Q) + C$ iterates, the growth lemma and the clean-magnet minorization split every surviving physical parent into finitely many disjoint legal collision cylinders. A subfamily of total conditional mass at least $\eta_{\text{reg}} > 0$ maps, with uniformly bounded distortion, onto the native children of Lemma 4.34 of [16]. For every $P \in \mathbb{R}_{d_Z}[z]$, the child deleted in Leclerc's cutting step has an actual collision subset of relative $\mu_{\beta,Q}^u$ -mass at least $\eta_{\text{cut}} > 0$ on which $|X_{\beta,Q} - P| > \sigma_Q^{1+\alpha_L}$. Endpoint overshoots have bounded multiplicity.*
- (PR3) *Repeating the clean-magnet trial on the complement gives $L_{j+1} \leq (1 - \eta_{\text{reg}})L_j + Ce^{-c(j+|\log \sigma_Q|)}$. All inverse branches and all buffer continuations are retained, and the total physical time of J trials and a depth- J Leclerc tree is $C(J + |\log \sigma_Q|)$.*

The minorization in (PR2) is on the actual collision-SRB standard family. It is not obtained by pushing a fractal equilibrium conditional through a smooth extension.

Proposition 6.22 (Single-source physical bracket). *Under $\text{SS}(d_Z)$, (46)–(47) hold on every labelled native interval. In the dominant coordinates there are positive labelled scales a_X, a_Y such that*

$$c \leq |X'|/a_X, |Y'|/a_Y \leq C, \quad \Xi = |\xi_1|a_X \asymp |\xi_2|a_Y. \quad (48)$$

Proof. The bracket assertion is precisely $\text{SS}(d_Z)$. The last display is the nonvanishing part of the clean-gate chart condition after contracted coordinates are removed. No uniform lower bound is asserted: the inverse scale cost is the explicit factor $\mathfrak{d}_\lambda$ in (69). $\square$

Lemma 6.23 (Gate, buffer, and native scale). *Let $\alpha_L < \alpha_B$, put*

$$\sigma = (r/c_\ell)^{1/(1+\alpha_L)}, \quad d_0 = \frac{\alpha_B - \alpha_L}{2}, \quad (49)$$

and call a context good when $H_\beta \leq \sigma^{-d_0}$. Choose $N = \lceil C_{\text{buf}} |\log r| \rceil$ so that $\kappa^N \leq r/10$. Then on good contexts the error in (47) is $o(r)$, while

$$\mathbb{P}\{H_\beta > \sigma^{-d_0}\} \leq \mathbb{E}(H_\beta^p) r^{pd_0/(1+\alpha_L)}. \quad (50)$$

The regeneration/cutting time from PR plus N is $O(|\log r|)$ and all buffer continuations are retained, so the refinement preserves conditional mass.

Proof. On a good context, $H_\beta \sigma^{1+\alpha_B} \leq \sigma^{1+(\alpha_B+\alpha_L)/2} = o(\sigma^{1+\alpha_L}) = o(r)$. Markov's inequality gives (50). The remaining claims follow from exponential gate hitting and the mass-preserving buffer. $\square$

Proposition 6.24 (Physical regeneration implies polynomial escape). *Under $\text{PR}(\alpha_L, d_Z)$ there is $\gamma_{\text{ph}} > 0$ such that, uniformly for $P \in \mathbb{R}_{d_Z}[z]$, every native interval obeys*

$$\mu_{\beta,Q}^u\{|X_{\beta,Q} - P| \leq \sigma_Q^{1+\alpha_L}\} \leq C\sigma_Q^{\gamma_{\text{ph}}}. \quad (51)$$

The same bound holds after multiplication by a positive marked amplitude, with the corresponding scaled amplitude mark.

Proof. Run the regeneration trials of (PR2)–(PR3) and, on every successful child, use Lemma 4.34 and Proposition 4.33 of [16]. At each cutting generation one Leclerc child of relative mass at least η_{cut} is disjoint from the polynomial sublevel set. Failed gate

trials lose a factor $1 - \eta_{\text{reg}}$, while endpoint overshoots and return tails contribute Ce^{-cj} . Consequently, after $J = \lfloor c_0 |\log \sigma_Q| \rfloor$ generations, the surviving physical mass is at most

$$C(1 - \eta_{\text{reg}}\eta_{\text{cut}})^J + Ce^{-cJ} \leq C\sigma_Q^{\gamma_{\text{ph}}}.$$

Bounded distortion and (PR1) compare every pulled-back child with its actual conditional mass. Multiplication by a positive amplitude is made before normalization and costs its scaled supremum. No horseshoe-equilibrium/collision-SRB comparison is used. $\square$

Theorem 6.25 (Physical single-source energy tree). *Let μ_β be the domain conditional and $\nu_\beta = (S_\beta)_*\mu_\beta$. With*

$$\Theta_E = \frac{\min\{\gamma_{\text{ph}}, pd_0\}}{1 + \alpha_L} > 0, \quad (52)$$

one has

$$\sum_\beta w_\beta \iint \mathbf{1}_{\{|S_\beta(t) - S_\beta(t')| \leq r\}} d\mu_\beta(t) d\mu_\beta(t') \leq Cr^{\Theta_E} + Ce^{-cN}, \quad (53)$$

$$\sup_z \sum_\beta w_\beta \nu_\beta([z - r, z + r]) \leq Cr^{\Theta_E/2} + Ce^{-cN}. \quad (54)$$

Proof. Fix t' and all frozen physical data. By (47), the event $|S_\beta(t) - S_\beta(t')| \leq r$ is contained, after enlarging constants, in $|X_{\beta,Q} - P_{\beta,Q,t'}| \leq \sigma^{1+\alpha_L}$ with $\sigma = (r/c_\ell)^{1/(1+\alpha_L)}$. On good contexts combine Proposition 6.22, Lemma 6.23, and (51). Equation (50) handles bad contexts, and the buffer/gate tails give e^{-cN} . Integrate in t' and then over branches; this order is essential because the bad-context estimate is averaged over the physical branch law and is not uniform for a fixed β . Finally use

$$\nu_\beta(I)^2 \leq (\nu_\beta \otimes \nu_\beta)\{|u - v| \leq 2|I|\}$$

and Jensen in the positive weights w_β to obtain (54). $\square$

Corollary 6.26 (Central slope small-ball). *The complete central collision-SRB quotient satisfies (54) with constants uniform in s .*

Proof. Apply Theorem 6.25 and the uniform marked return/grazing/context moments. $\square$

Lemma 6.27 (Slope small-ball controls the balanced phase). *For $\psi_\xi = \xi_1 X + \xi_2 Y$ on an affine-normalized gate chart, let*

$$\Xi = |\xi_1|a_X \asymp |\xi_2|a_Y, \quad (55)$$

with a_X, a_Y as in (48). The mass of $\{|\psi'_\xi| \leq \Xi r\}$ is at most $C(r^{\Theta/2} + e^{-cn})$. Smooth value-space cutoffs on its complement have C^α norm $O(r^{-1})$ and satisfy $\inf |\psi'_\xi| \geq c\Xi r$.

Proof. By Proposition 6.22,

$$\psi'_\xi = X'(\xi_1 + \varepsilon_{\text{or}}\xi_2 e^S).$$

The lower and upper bounds in (48) convert the bracketed factor to the physical scale (55). Only opposite-oriented coefficients can resonate, and the resonant set is contained in one interval $|S - z_\xi| \leq Cr$. Apply (54). Composing a fixed smooth three-term partition with $(\xi_1 + \varepsilon_{\text{or}}\xi_2 e^S)/(\Xi r)$ gives the cutoff and norm bounds. $\square$

Lemma 6.28 (Global double projection before disintegration). *Let $\mathcal{G}_\alpha$ be the projective space of signed graph currents with the explicit norm*

$$\|J\|_{\mathcal{G}_\alpha} = \inf_{J=\sum_\lambda \epsilon_\lambda J_\lambda} \sum_\lambda |J_\lambda|(1)\{1 + \text{Mark}(\lambda) + Z_q(\lambda) + (1 + [S_\lambda]_{C^\alpha})^{p_F}\}, \quad (56)$$

and weak total-variation norm $\|\cdot\|_{\mathcal{G}_{\alpha,w}}$. Let $\mathcal{M}_\alpha^\pm$ be the corresponding one-sided marginal spaces, and let $\mathcal{R}_\alpha$ be the regular product-current space. Work in the direct-sum space

$$\mathfrak{C}_\alpha = \mathcal{G}_\alpha \oplus_\pi (\mathcal{M}_\alpha^- \hat{\otimes} \nu^+) \oplus_\pi (\nu^- \hat{\otimes} \mathcal{M}_\alpha^+) \oplus_\pi \mathcal{R}_\alpha, \quad (57)$$

where ν^-, ν^+ are fixed smooth probability references. Let J be the complete signed face current in common physical coordinates. If m_J^-, m_J^+ are its marginals, define $P_- J = m_J^- \otimes \nu^+$ and $P_+ J = \nu^- \otimes m_J^+$. Put

$$J^\circ = (I - P_-)(I - P_+)J = J - m_J^- \otimes \nu^+ - \nu^- \otimes m_J^+ + J(1,1)\nu^- \otimes \nu^+. \quad (58)$$

Then J° is double centered. If the complete response current already has zero marginals, then

$$J^\circ = J. \quad (59)$$

Only after this identity is formed in physical coordinates do we disintegrate the graph component; the two rank-one components and the regular product component remain separate blocks of (57). If the original physical amplitudes obey (100) and the references have fixed C^α densities, then

$$\|J^\circ\|_{\mathfrak{C}_\alpha} \leq C\|J\|_{\mathcal{G}_\alpha}, \quad \sum_\lambda \|J_{\lambda,\text{graph}}^\circ\|_{\mathcal{G}_\alpha} + \|J_{\text{rank}}^\circ\| + \|J_{\text{reg}}^\circ\| < \infty. \quad (60)$$

Proof. The operators commute and $P_- P_+ J = J(1,1)\nu^- \otimes \nu^+$, proving the algebraic formula and centering. Marginalization of one normalized graph piece is bounded from $\mathcal{G}_\alpha$ to $\mathcal{M}_\alpha^\pm$ by Fubini, the scaled C^α norm, and the upper/lower conditional-density bounds. Tensoring with a fixed smooth probability is bounded into the corresponding rank-one summand of (57). It is *not* asserted to produce another graph current. The regular two-dimensional Jacobian/density derivative is placed in $\mathcal{R}_\alpha$. Taking the infimum over projective decompositions and using (100) proves (60). Only the surviving graph summand is refined by face labels; hence no cancellation is postponed until after label-dependent coordinates are chosen. $\square$

Lemma 6.29 (Dominant-coordinate reduction). *After assembly and the recentering of Lemma 6.28, retain the rank-one and regular-product summands in the projective tensor space and evolve both factors by their centered operators. On a clean graph rectangle in the diagonal band $D^{-1}M < N < AM$, the graph-current pairing decomposes as*

$$B_{M,N} = B_{M,N}^{\text{dom,dom}} + B_{M,N}^{\text{dom,rem}} + B_{M,N}^{\text{rem,dom}} + B_{M,N}^{\text{rem,rem}}. \quad (61)$$

The first term is a one-variable integral with physical coordinates X, Y . Every other term is bounded by

$$Ce^{-c \max\{M,N\}} \|g\|_{C^1} \|f\|_{C^1}. \quad (62)$$

The global joint constant and axis blocks vanish or remain in the explicitly separated rank-one summands before the graph-rectangle sum is estimated. Every graph-axis mode created by a boundary cutoff is uniformly nonstationary and is charged to its cutoff norm.

Proof. In stable–unstable endpoint coordinates, freeze the coordinate contracted by the relevant inverse return and subtract the frozen value. The mean-value theorem and hyperbolicity give remainders bounded by $C\lambda_-^M \|g\|_{C^1}$ and $C\lambda_+^N \|f\|_{C^1}$. In the diagonal band these imply (62). The two frozen functions depend only on X, Y , proving (61). Global projection separates the constant and two axis blocks before any label-dependent chart is chosen. They are estimated in their rank-one or regular-product tensor spaces, not redescribed as graph currents; Lemma 6.32 supplies product decay. A graph boundary cutoff can reintroduce an axis term, but it has phase $\xi_1 X$ or $\xi_2 Y$; (48) gives the physical derivative lower bound, and its cutoff ledger is absolutely summable by (60). $\square$

Proposition 6.30 (Scaled physical face density). *On a homogeneous physical face cell $Q = t_Q + \ell_Q[-1, 1]$, the collision–SRB conditional is $d\mu_Q^u = \rho_Q(t) dt$, where, before normalization,*

$$\rho_Q^{\text{raw}}(t) = c_\mu \cos \varphi(t) J_{\text{coa}}(t) J_{\text{hol}}(t) J_{\text{chart}}(t), \quad J_{\text{coa}} = |\nabla G|^{-1}. \quad (63)$$

Put $\hat{\rho}_Q(u) = \ell_Q \rho_Q(t_Q + \ell_Q u)$. There is a mark

$$\text{Mark}(Q) = 1 + \mathfrak{h}_Q + \mathfrak{b}_Q + [\log J_{\text{hol}}]_{\alpha, Q} + [\log J_{\text{coa}}]_{\alpha, Q} + [\log J_{\text{chart}}]_{\alpha, Q} + \|w_Q\|_{\mathcal{A}^\alpha} \quad (64)$$

such that

$$C^{-1} \text{Mark}(Q)^{-1} \leq \hat{\rho}_Q \leq C \text{Mark}(Q), \quad [\log \hat{\rho}_Q]_\alpha \leq C \text{Mark}(Q). \quad (65)$$

For some $p_d > 2$,

$$\sum_Q M_Q \text{Mark}(Q)^{p_d} < \infty \quad (66)$$

uniformly in the small table family. The same statement holds after the positive/negative decomposition of a signed face amplitude.

Proof. In collision coordinates the invariant density is $c_\mu \cos \varphi dr d\varphi$. Coarea along the physical graph, followed by stable projection to the gate coordinate, gives (63). On a homogeneous strip, $\cos \varphi$, the chart Jacobian, and the holonomy Jacobian have the standard scaled log–Hölder distortion. Dividing by $\int_Q \rho_Q^{\text{raw}}$ gives (65); the factor ℓ_Q removes the cell scale.

The homogeneity contribution has a fixed polynomial cost, while the strip mass is exponentially summable. The holonomy and chart contributions are summed by the context moment, and the amplitude contribution by the marked face ledger. Generalized Hölder with slightly smaller exponents proves (66). For a signed amplitude q , write $q = (2\|q\|_\infty + q) - 2\|q\|_\infty$; multiplication by the positive masses restores the scaled norm and introduces no reciprocal small-mass factor. $\square$

Lemma 6.31 (Fractional nonstationary phase). *Let I be an interval, $a \in C^\alpha(I)$, and $\phi \in C^{1+\alpha}(I)$. On each component of the support of a , suppose ϕ' has one sign and $|\phi'| \geq \lambda > 0$. Put $\Phi = \sum_C \int_C |\phi'(t)| dt$, the total phase length of the support. For $C = (l_C, r_C)$ define the endpoint functional $\mathfrak{E}(a) = \sum_C (|a(l_C+)| + |a(r_C-)|)$.*

$$\left| \int_I e^{i\phi(t)} a(t) dt \right| \leq C_\alpha \left\{ \frac{\mathfrak{E}(a)}{\lambda} + \frac{\Phi[a]_\alpha}{\lambda^{1+\alpha}} + \frac{\Phi\|a\|_\infty[\phi']_\alpha}{\lambda^{2+\alpha}} \right\}. \quad (67)$$

The estimate adds over a countable union of sign components without a component-count loss.

Proof. On one sign component use $y = \phi(t)$ and write $b(y) = a(\phi^{-1}y)/\phi'(\phi^{-1}y)$. Then

$$\|b\|_\infty \leq \|a\|_\infty/\lambda, \quad [b]_\alpha \leq [a]_\alpha \lambda^{-1-\alpha} + \|a\|_\infty [\phi']_\alpha \lambda^{-2-\alpha}.$$

For a shift $h = \pi$, the identity

$$2 \int e^{iy} b(y) dy = \int e^{iy} \{b(y) - b(y+h)\} dy + \text{endpoint terms}$$

gives $C\Phi[b]_\alpha$ plus the two endpoint values of b on that component. Since $|b| \leq |a|/\lambda$, summing the endpoint values gives $\mathfrak{E}(a)/\lambda$, while the sum of the component y -lengths is Φ . This proves (67); no number-of-components estimate is hidden in the supremum term. $\square$

Lemma 6.32 (Rank-one product under double centering). *If $\eta_\pm$ are admissible dual collision currents and $R = \eta_- \otimes \eta_+$, then*

$$R(Q_-^m g \otimes Q_+^n f) = \eta_-(Q_-^m g) \eta_+(Q_+^n f).$$

Thus the two one-sided gaps give $C\tau^m \vartheta^n$.

Proof. This is the universal property of the projective tensor product followed by the two admissible dual bounds. In the principal proof the rank-one corrections are instead included in the low-frequency functional of Proposition 6.37; this lemma is only an algebraic consistency check. $\square$

Proposition 6.33 (Central balanced estimate with explicit budget). *Let Θ be the energy exponent and set*

$$A_{\text{frac}} = 3 + \alpha, \quad 0 < \delta < \frac{\alpha}{2A_{\text{frac}}}, \quad \beta_{\text{cen}} = \min\{\delta\Theta/2, \alpha - A_{\text{frac}}\delta\} > 0.$$

Then every balanced nonzero graph mode has saving $C\Xi^{-\beta_{\text{cen}}}$ times a fixed power of $\text{Mark}(Q)$. If the observable regularity satisfies $r_{\text{obs}} > 2 + \beta_{\text{cen}}$, the complete lattice sum is exponentially small in the two induced depths. Low graph modes, rank-one corrections, and the regular product block are treated together by Proposition 6.37.

Proof. Take $r = \Xi^{-\delta}$. The energy theorem bounds the resonant mass by $C\Xi^{-\delta\Theta/2} + Ce^{-cN}$. On its complement use two smooth cutoffs in $\psi'_\xi/(\Xi r)$, one for each sign. They vanish on the resonant boundaries. Sum adjacent physical cells before taking absolute values, so their artificial endpoint terms cancel with opposite orientation. The remaining face, singularity, and homogeneity endpoints obey

$$\sum_Q \mathfrak{E}(a_Q) \leq Cr^{-1} \sum_Q M_Q \text{Mark}(Q) \{1 + Z(\mathcal{G}_Q)\}, \quad (68)$$

by Proposition 6.42. Proposition 6.30 gives

$$\|a_\pm\|_\infty + [a_\pm]_\alpha \leq C \text{Mark}(Q) r^{-1}.$$

Furthermore

$$\lambda = c\Xi r, \quad \Phi \leq C\Xi, \quad [\psi'_\xi]_\alpha \leq C\Xi \text{Mark}(Q).$$

Lemma 6.31 therefore gives

$$\begin{aligned} |\mathcal{I}_Q(\xi)| &\leq C \text{Mark}(Q)^2 \{(1 + Z(\mathcal{G}_Q))\Xi^{-1}r^{-2} + \Xi^{-\alpha}r^{-(3+\alpha)}\} \\ &\leq C \text{Mark}(Q)^2 (1 + Z(\mathcal{G}_Q))\Xi^{-\beta_{\text{cen}}}. \end{aligned}$$

The joint estimate (82) sums the cells without separating the density and boundary moments.

For completeness, put $K = e^{\eta N}$ with $0 < \eta < c_{\text{FC}}/(2A_{\text{dom}})$. After global centering and the exact dominant-coordinate reduction, the face-adapted projector $\mathbf{S}_K^{\text{dom}}$ retains precisely the scalar modes satisfying $|\xi_1|a_X \leq K$ and $|\xi_2|a_Y \leq K$. Proposition 6.37 treats this finite sum with loss $K^{A_{\text{dom}}}$. Every balanced mode of the complementary dominant sum has $\Xi > K$ by definition, so the preceding saving is $e^{-\eta\beta_{\text{cen}}N}$. Contracted-coordinate terms were removed before the cutoff and are exponentially small by Lemma 6.29; unbalanced scalar modes are uniformly nonstationary. The two-dimensional dominant lattice sum converges because $r_{\text{obs}} > 2 + \beta_{\text{cen}}$. Lemma 6.23 supplies the resolving depth. Global marginal corrections are inserted into the same finite low-frequency functional before any absolute value is taken. The regular two-dimensional density is not placed in the face lattice: Corollary 6.38 uses its independent $W^{R,1}$ smoothing tail. $\square$

6.11 Low frequencies after dominant-coordinate reduction

The cutoff is imposed only after the graph term has been reduced to its two scalar dominant coordinates. This avoids the false implication that a large ambient Fourier vector must have a large restriction to a one-dimensional face. Contracted coordinates are estimated before the cutoff.

Definition 6.34 (Dominant event norm and separate dynamic-Hölder norm). *Fix a finite physical product atlas and assemble coincident artificial faces before inserting its partition of unity. For a face label λ let $a_{X,\lambda}, a_{Y,\lambda} > 0$ be the two scales in (48) and put*

$$\mathfrak{d}_\lambda = 1 + a_{X,\lambda}^{-1} + a_{Y,\lambda}^{-1}.$$

Write

$$\mathcal{K}_{\text{ev}} = k_{\text{reg}} dm_{M \times M} + \sum_{\lambda} M_{\lambda}(H_{\lambda})_*(w_{\lambda} d\bar{s})$$

and, for an integer $1 \leq R \leq 3$ allowed by the C^3 table regularity, define

$$\|\mathcal{K}_{\text{ev}}\|_{\text{phys},R} := \sum_{\chi} \|\zeta_{\chi} k_{\text{reg}}\|_{W^{R,1}} + \sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{\mathcal{A}^{\alpha}} \mathfrak{d}_{\lambda}^3 \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_{\lambda}^{-}) + Z(\mathcal{G}_{\lambda}^{+})\}. \quad (69)$$

For bounded $F : M^q \rightarrow \mathbb{C}$ let

$$\|F\|_{\text{SH}} := \|F\|_{\infty} + \sum_{i=1}^q \sup_{x_j: j \neq i} \sup_{x_i \neq y_i} \frac{|F(\dots, x_i, \dots) - F(\dots, y_i, \dots)|}{\theta_0^{s_i(x_i, y_i)}}. \quad (70)$$

The stable or unstable separation time appropriate to the i th slot is used, as in [5, Definition 2.2].

On a localized graph label, perform Lemma 6.29 first. Periodize the two scalar dominant coordinates and retain the integer pairs satisfying

$$|\xi_1|a_{X,\lambda} \leq K, \quad |\xi_2|a_{Y,\lambda} \leq K. \quad (71)$$

Let $A_{\lambda,K}^{\text{dom}}$ be the resulting finite Fourier sum, extended smoothly inside the fixed product chart. On the regular density use the ordinary four-dimensional convolution $\rho_K^{(4)} * k_{\text{reg}}$. With $\mathbf{D} = (I - P_-)(I - P_+)$ define

$$\mathbf{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}} := \mathbf{D} \left(\left[\rho_K^{(4)} * k_{\text{reg}} + \sum_{\lambda} M_{\lambda} A_{\lambda,K}^{\text{dom}} \right] dm_{M \times M} \right), \quad \mathbf{H}_K^{\text{dom}} = I - \mathbf{S}_K^{\text{dom}}. \quad (72)$$

All artificial endpoints are assembled before (72); marginal corrections created by $\mathbf{D}$ remain in the displayed finite sum. In the sequel the smoothed current is identified with its bounded density when the SH norm is used.

Lemma 6.35 (Polynomial cost of the dominant truncated event). *Put $A_{\text{dom}} = 5$. For $K \geq 2$,*

$$\|\mathbf{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}}\|_{\text{SH}} \leq CK^{A_{\text{dom}}} \|\mathcal{K}_{\text{ev}}\|_{\text{phys}, \mathbf{R}}. \quad (73)$$

The constant is independent of return and grazing truncations.

Proof. For one face label the number of pairs in (71) is at most $CK^2 \mathfrak{d}_\lambda^2$ and one separate difference costs at most $CK \mathfrak{d}_\lambda$. Thus its cost is $CK^3 \mathfrak{d}_\lambda^3$. Hyperbolicity converts ordinary coordinate differences to the stable/unstable separation-time bounds required by [5, Theorems 2.3–2.4]. Sum with (69). The regular four-dimensional convolution has $O(K^4)$ modes and one derivative, hence cost K^5 . The fixed atlas, extensions, partition multipliers, and $\mathbf{D}$ are bounded. Taking the larger exponent gives $A_{\text{dom}} = 5$. $\square$

Lemma 6.36 (Regular high-frequency tail). *For the regular two-dimensional density,*

$$\|(I - \rho_K^{(4)} *) k_{\text{reg}}\|_{L^1(M \times M)} \leq CK^{-R} \|k_{\text{reg}}\|_{W^{R,1}(M \times M)}. \quad (74)$$

Consequently its pairing with $Q^m g$ and $Q^n f$ is bounded by the right-hand side times $\|g\|_\infty \|f\|_\infty$, uniformly in m, n .

Proof. This is the standard approximation-of-the-identity estimate in L^1 ; Markov normalization gives $\|Q^m h\|_\infty \leq 2\|h\|_\infty$. $\square$

Proposition 6.37 (Central low-frequency estimate). *Let $f^\circ = f - \mu_0(f)$ and $g^\circ = g - \mu_0(g)$. In the diagonal band put $N = \max\{m, n\}$ and $K = e^{\eta N}$. Then*

$$\left| \left\langle Q^n (\mathbf{S}_K^{\text{dom}} K_V) Q^m g, f \right\rangle \right| \leq C e^{-(c_{\text{FC}} - A_{\text{dom}} \eta) N} \|g\|_{C^r} \|f\|_{C^r}, \quad (75)$$

provided $0 < \eta < c_{\text{FC}}/(2A_{\text{dom}})$.

Proof. The finite sum is an absolutely convergent sum of expectations of

$$FK(x_{-m}, x_0, x_1, x_{n+1}) = g^\circ(x_{-m}) A_K(x_0, x_1) f^\circ(x_{n+1}).$$

Split first after x_{-m} . Its product term vanishes by centering of g and [5, Theorem 2.3] gives $CK^{A_{\text{dom}}} \theta_{\text{FC}}^m$. Splitting instead before x_{n+1} gives the analogous bound with n . Taking the better estimate gives $CK^{A_{\text{dom}}} \theta_{\text{FC}}^N$; the dominant event norm sums all labels and all marginal corrections. $\square$

Corollary 6.38 (Regular defect via functional correlation). *The regular part of K_V has central-band exponential decay: its low part is covered by Proposition 6.37, and Lemma 6.36 gives $K^{-R} = e^{-\eta R N}$ for its high part. No tensor, wavelet, or face-phase estimate is used.*

6.12 Radial faces: direct flux boundary control

We retain every connected component. Let $\mathcal{W}_\lambda(j, k) = \{W_{\lambda, j, k, r}\}_r$ be the maximal components of one coarea face with incoming strip j and outgoing strip k . Put

$$p_{\lambda, j, k, r} = \int_{W_{\lambda, j, k, r}} \rho_\lambda^{\text{raw}} ds, \quad M_\lambda = \sum_{j, k, r} p_{\lambda, j, k, r}, \quad \bar{p}_{\lambda, j, k, r} = p_{\lambda, j, k, r} / M_\lambda.$$

Thus $\bar{p}$, not p , is the weight of the normalized standard family.

Lemma 6.39 (Cone-oriented monotonic component bound). *Remove the pieces whose tangent has angle at most ϵ with both usable cones. Cut every remaining face at changes of its chosen cone orientation. On each resulting chart, one of the forward or reversed impact coordinates is strictly monotone. Hence, for fixed incoming/outgoing homogeneity ranks (j, k) , there are at most $4N_{\text{cone}}(\lambda)$ connected components and*

$$\sum_{j,k,r} \frac{1}{(1+j)^2(1+k)^2} \leq CN_{\text{cone}}(\lambda). \quad (76)$$

The label moment of N_{cone} is absorbed by the return/grazing ledger.

Proof. In an oriented collision chart the impact-coordinate derivative has one sign; for the vertical model it is exactly $dr_1/d\varphi = -\tau/\cos\varphi_1$, as in [14, Lemma 5.1]. Composition with a cone-transverse face chart preserves the sign in one of the two time orientations. Each homogeneity strip has two sides, so a monotone chart meets a fixed pair of strips in at most four maximal intervals. Sum over the $N_{\text{cone}}(\lambda)$ charts and then over j, k . The finite-depth pathwise jet condition controls the number of orientation changes, while exponential return/grazing tails sum its label mark. $\square$

Lemma 6.40 (Two-flux coarea Jacobian). *On every cone-oriented face component the unnormalized arclength density is*

$$\rho_\lambda^{\text{raw}} = \frac{|a_\lambda|}{|\nabla G_\lambda|} J_{\text{chart},\lambda} J_{\text{hol},\lambda} \frac{\cos\varphi_- \cos\varphi_+}{\tau}. \quad (77)$$

The formula is invariant under switching to the reversed orientation.

Proof. Arclength coarea contributes $|a_\lambda|/|\nabla G_\lambda|$. In collision coordinates the SRB flux through the incoming section is $c_\mu \cos\varphi_- dr_-$. The billiard differential gives $dr_+/d\varphi_- = -\tau/\cos\varphi_+$; changing to outgoing arclength therefore contributes $\cos\varphi_+/\tau$. The fixed chart and stable-holonomy changes contribute J_{chart} and J_{hol} . Multiplying these four exact factors proves the formula. Time reversal interchanges the two cosine factors and preserves their product. $\square$

Definition 6.41 (Flux-complexity condition FZ). *A face family satisfies $\text{FZ}(\theta_Z)$ if the density of Lemma 6.40 has marks $\mathfrak{M}_\lambda, \mathfrak{C}_\lambda \geq 1$ such that*

$$p_{\lambda,j,k,r} \leq C\mathfrak{M}_\lambda \frac{|W_{\lambda,j,k,r}|}{(1+j)^2(1+k)^2}, \quad (78)$$

$$\sum_{j,k,r} \frac{1}{(1+j)^2(1+k)^2} \leq \mathfrak{C}_\lambda := CN_{\text{cone}}(\lambda), \quad (79)$$

and, for some $q_F > 1$,

$$\sum_\lambda M_\lambda \left[\text{Mark}(\lambda)^2 \left\{ 1 + \frac{\mathfrak{M}_\lambda \mathfrak{C}_\lambda}{M_\lambda} \right\} \right]^{q_F} < \infty. \quad (80)$$

Pieces whose tangent makes angle at most ϵ with both usable cones have total marked mass at most $C\epsilon^{\theta_Z}$. This is the required near-stable angle-sublevel input. On the complementary good-angle pieces, the product of all inverse-angle costs in the graph chart, density, nonstationary phase, and recovery maps is at most $C\epsilon^{-A_{\text{ang}}}$ for a fixed $A_{\text{ang}} > 0$. Both constants are part of FZ.

Proposition 6.42 (Flux-weighted radial boundary lemma). *Assume $\text{FZ}(\theta_Z)$ and decompose the signed amplitude into positive pieces. Then the normalized standard family obeys*

$$M_\lambda Z(\mathcal{G}_\lambda) = \sum_{j,k,r} \frac{p_{\lambda,j,k,r}}{|W_{\lambda,j,k,r}|} \leq C \mathfrak{M}_\lambda \mathfrak{E}_\lambda, \quad (81)$$

and, for the actual physical labels,

$$\sum_\lambda M_\lambda \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\} < \infty. \quad (82)$$

and the near-stable pieces have the marked ϵ^{θ_Z} tail in Definition 6.41.

Proof. Divide (78) by the component length and sum over the full triple (j, k, r) . Equation (79) gives (81); no component is suppressed. Multiply by the squared density/amplitude mark and use (80) (first with power one) to obtain (82). The exact two-flux Jacobian, weighted component complexity, and angle sublevel are the three falsifiable geometric inputs; none is deduced merely from the homogeneity labels. $\square$

Lemma 6.43 (Explicit angle balance). *Suppose the good-angle core of a central or off-diagonal estimate is $Ce^{-c_{\text{core}}N}$ before inverse-angle loss. Set*

$$\epsilon_N = e^{-\zeta N}, \quad \zeta = \frac{c_{\text{core}}}{A_{\text{ang}} + \theta_Z}.$$

Then the sum of the bad-angle and good-angle contributions is at most $Ce^{-c_Z N}$, where

$$c_Z = \frac{c_{\text{core}}\theta_Z}{A_{\text{ang}} + \theta_Z} > 0. \quad (83)$$

Proof. The bad-angle mass is $e^{-\zeta\theta_Z N}$, while the good-angle estimate is $e^{-(c_{\text{core}} - A_{\text{ang}}\zeta)N}$. The displayed choice makes the two exponents equal. $\square$

Corollary 6.44 (Uniform radial recovery). *For each time orientation, the radial family is regular after at most*

$$R^\pm \leq \alpha_1 \log\{1 + Z(\mathcal{G}_{\text{rad}}^\pm)\} + \alpha_2 \quad (84)$$

iterates. After multiplication by physical mass, the recovery-window sum in all four time quadrants is finite and is controlled by (82).

Proof. Apply [14, Corollary 6.15(c)] to the positive standard families in each orientation. Use $\log(1 + Z) \leq C_\epsilon(1 + Z)^\epsilon$ and multiply by M_λ . The joint mark- Z ledger then sums both orientations and all pre-recovery iterates. This is the weighted hypothesis of Proposition 4.1. $\square$

Lemma 6.45 (Observable multiplier to positive proper families). *Let (W, ρ) be a positive standard pair and let $q_m = (g \circ T^{-m})\rho$ on the time orientation for which W is unstable. There is $C_m \leq C\|g\|_{C^\alpha}$ such that*

$$q_m = (q_m + C_m \rho) - C_m \rho$$

is a difference of positive densities. Their total masses are bounded by $C\|g\|_{C^\alpha}$ and their scaled log-Hölder marks are at most $Ce^{x^m} \text{Mark}(W, \rho)$. After

$$R_m \leq C_0 + C_1 m + C_2 \log\{1 + Z(\mathcal{G})\} \quad (85)$$

iterates, both positive families are proper. The reversed collision map has the analogous statement.

Proof. Take $C_m = 2\|g\|_\infty$. Positivity and the mass bound are immediate. Hyperbolicity and bounded distortion give $[g \circ T^{-m}]_{\alpha, \text{sc}} \leq Ce^{\lambda m} \|g\|_{C^\alpha}$. Since the added density is at least $C_m \rho/2$, division by it gives the same scaled log-Hölder bound. The standard growth lemma first absorbs this mark in $C_1 m$ iterates; Corollary 6.15(c) of [14] absorbs the boundary functional in $C_2 \log(1 + Z)$ further iterates. Time reversal gives the last assertion. $\square$

Proposition 6.46 (Signed proper-family mapping and off-diagonal bounds). *Assume the forward and reversed billiard growth/coupling theorem on positive proper standard families, $\text{FZ}(\theta_Z)$, and the scaled density bound of Proposition 6.30. Decompose every signed face amplitude as the difference of two positive amplitudes before normalization. Then there are positive exponents $\alpha_\pm, \beta_\pm$ such that the complete face current satisfies*

$$|B_{m,n}| \leq Ce^{\beta_- m - \alpha_+ n}, \quad (86)$$

$$|B_{m,n}| \leq Ce^{-\alpha_- m + \beta_+ n}. \quad (87)$$

The constants include all recovery windows and are uniform under the physical and flux ledgers.

Proof. Apply Lemma 6.45 on each positive face piece. Its recovery time is bounded by (85). Equations (81) and (84) control the boundary part of that recovery. Coupling for the remaining $n - R_\lambda^+$ iterates gives

$$CM_\lambda \text{Mark}(\lambda)^2 (1 + Z_\lambda^+) e^{\beta_- m - \alpha_+ (n - R_\lambda^+)_+}.$$

Before recovery use total variation. The inequality $e^{\alpha_+ R_\lambda^+} \leq Ce^{\alpha_+ C_1 m} (1 + Z_\lambda^+)^{\alpha_+ C_2}$ and (82), with a weakened coupling exponent chosen below the available mark- Z moment, sum all labels and recovery windows. Absorb $e^{\alpha_+ C_1 m}$ into $e^{\beta_- m}$. This is (86). Apply the same argument to the reversed collision map for (87); positive/negative pieces are recombined only after the two bounds are established. $\square$

7 Slope-energy and fractional-phase proof architecture

The proof now has six noncircular links.

- (F1) Quotient the exponential Young return on the regular magnet to the one-state countable $C^{1+\alpha}$ IFS of Proposition 6.10. Physical side labels remain in the signed amplitude fiber.
- (F2) Lemma 6.17 combines the exponential return tail with the maximal grazing rank and proves the joint moment needed for the countable quotient and the marked coarea amplitudes.
- (F3) The normal-cycle value $\mathfrak{A} = -1$ motivates the clean template; SS and the actual-SRB regeneration PR, together with the mass-preserving buffer, give the energy theorem and Jensen.
- (F4) Proposition 6.22 and smooth value-space cutoffs and Lemma 6.31 give the balanced Fourier estimate without curvature or componentwise phase partitions.
- (F5) Global centering is performed before the physical chart sum. Lemma 6.29 removes contracted modes; Proposition 6.37 treats the remaining low modes, while Proposition 6.33 treats the high modes.

- (F6) At each cutoff coarea currents converge at fixed depth. The component-indexed FZ estimate, QT_{path} , and the strict scalar-tail margin remove L, H before recovery summation.

This order makes clear which external results are used. Leclerc’s Lemma 3.15 supplies the cohomological bracket and his Propositions 4.31–4.33 supply the four-point all-scale polynomial escape. The final oscillatory estimate is the elementary fractional phase lemma proved above; LPS is retained only as background for the quotient route. Bertozzi’s triple inducing supplies a useful model for the one-state gate architecture, not the slope conclusion. The positive entrance/exit comparison is the symbolic local product argument of [24]. The one-step expansion, linear singularity complexity, and uniform billiard constants used by the master atlas are summarized in [2].

The route deliberately does not use the recent curvilinear two-ends theorem or L^2 flattening: their nonlinear applications require C^2 or higher geometry, whereas the billiard quotient available here is only $C^{1+\alpha}$. It also does not infer a product tower tail from two marginal Chernoff bounds: the diagonal product exponent comes from time comparability and functional correlation, while the off-diagonal regions are handled before the three-region assembly.

7.1 Execution audit

The central analytic chain is now represented by explicit propositions. The remaining primitive geometric inputs are explicitly named SS, PR, FZ, and pathwise QT. The exact gate interface is Lemma 6.18, Proposition 6.19, and Lemma 6.23; the slope-energy theorem is Theorem 6.25; the balanced phase is Proposition 6.33; and the low-frequency collision-time estimate is Proposition 6.37. Radial recovery is applied only after these estimates through Corollary 6.44 and Proposition 4.1.

7.2 A quantitative smoothing test

There is one falsifiable way to obtain (16). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (88)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (89)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad \Lambda_+ e^{-a\alpha} < 1. \quad (90)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (89) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (89). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 7.1 (Uniform difference-quotient closure). *Suppose either (14)–(15), or the shifted estimate (11) and the summability condition on the right of (12), hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$. If each block converges weakly to $K_{V,b}$, then the parameter difference quotient of the prescribed susceptibility has a summable majorant and converges term by term to the double series generated by K_V .*

Proof. In the first case apply Theorem 3.2; in the second apply Proposition 4.1. Either hypothesis gives an integrable majorant after summing over b, m, n . Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

For the moving table, the uniform hypothesis in this proposition can itself be reduced to four checkable pieces.

Lemma 7.2 (Parameter derivatives of contracting return words). *Let $h_{j,s}$ be the inverse one-step branches in a fixed compact regular atlas and assume, uniformly in j and $|s| < s_0$,*

$$\|Dh_{j,s}\|_\infty \leq \lambda < 1, \quad \|\partial_s h_{j,s}\|_{C^1} + \|\partial_s \log |h'_{j,s}|\|_\infty \leq A.$$

For a clean word $a = (a_1, \dots, a_R)$ let $h_{a,s} = h_{a_1,s} \circ \dots \circ h_{a_R,s}$. Then

$$\|\partial_s h_{a,s}\|_\infty \leq \frac{A}{1-\lambda}, \quad \|\partial_s \log |h'_{a,s}|\|_\infty \leq C(1+R). \quad (91)$$

The same linear-in- R bound holds for products of uniformly C^1 Jacobian, stable-holonomy, and coarea factors. Hence an exponential return moment dominates the difference quotients of all clean marked amplitudes and their return tail.

Proof. Differentiate the composition. The contribution created at position k is multiplied on the left by at least $k-1$ contracting derivatives, so its C^0 norm is bounded by $A\lambda^{k-1}$; summation gives the first bound. The logarithmic derivative of a composition is a sum of R one-step logarithmic derivatives. Differentiating it gives a uniformly bounded direct term at every position and orbit-displacement terms controlled by the first bound and uniform C^2 distortion. This gives $C(1+R)$. The product statement follows after taking logarithms. Finally $R^p e^{-\epsilon R}$ is summable for every fixed p , so the exponential return moment absorbs all these factors. $\square$

This lemma shows that long return words do not create an exponential parameter-derivative loss on the inverse-branch side. The remaining issue is not the clean word calculus but the common refinement of words that approach or cross a moving singularity; those pieces must be charged to the already constructed face/recovery tree rather than differentiated as clean branches.

That singular refinement is most naturally made before fixing the parameter.

7.3 Parameter-state geometry and cutoff removal

Definition 7.3 (Pathwise quantitative jet-tail condition). *Let $a(s)$ be the chosen analytic or circular table deformation with $a(0) = a_0$. For every face pair of depth at most L , let $\Delta_{\lambda\lambda'}(s)$ be a finite collection of jet minors covering the compact common-zero set after coincident germs are grouped. The path satisfies QT_{path} if there are $A, B, c_{\text{lab}}, A_{\text{loss}}, c_{\text{tail}} > 0$ such that*

$$|\Delta_{\lambda\lambda'}(0)| \geq e^{-AL}, \quad \sup_{|s| < \bar{s}} |\partial_s \Delta_{\lambda\lambda'}(s)| \leq Ce^{BL}, \quad (92)$$

$$\#\{(\lambda, \lambda') : |\lambda| \leq L\} \leq Ce^{c_{\text{lab}}L}. \quad (93)$$

All inverse-angle, coarea, chart, density, and parameter-derivative losses on the depth- L good set are at most $Ce^{A_{\text{loss}}L}$, while the unsmoothed return/grazing mass is at most $Ce^{-c_{\text{tail}}L}$. The strict path margin is

$$c_{\text{tail}} - A_{\text{loss}} > A + B. \quad (94)$$

For a finite-dimensional ambient family, the sublevel estimate

$$\text{Leb}\{a : |\Delta_{\lambda\lambda'}(a)| < e^{-AL}\} \leq C e^{-A\tau L}, \quad A\tau > c_{\text{lab}},$$

is a sufficient Borel–Cantelli mechanism for the first inequality at almost every base parameter, but it is not a substitute for the derivative bound or the path margin.

Lemma 7.4 (Persistence up to a depth depending on the parameter). *Choose*

$$\frac{1}{c_{\text{tail}} - A_{\text{loss}}} < \kappa < \frac{1}{A + B}, \quad L(s) = \lfloor \kappa \log(1/|s|) \rfloor. \quad (95)$$

For all sufficiently small s and all $L \leq L(s)$,

$$|\Delta_{\lambda\lambda'}(\theta s)| \geq \frac{1}{2} e^{-AL}, \quad 0 \leq \theta \leq 1.$$

Thus every noncoincident depth- $L(s)$ face family remains transverse along the whole coarea segment from 0 to s .

Proof. The mean-value theorem and (92) give

$$|\Delta(\theta s) - \Delta(0)| \leq C|s|e^{BL}.$$

For $L \leq L(s)$ the ratio of this error to e^{-AL} is at most $C|s|^{1-\kappa(A+B)} = o(1)$. The interval for κ is nonempty precisely because of (94). $\square$

Proposition 7.5 (Depth-dependent coarea and pathwise scalar tail). *Assume $|\nabla_x G_\lambda| \geq c_G > 0$ on every active face. For all depths $L \leq L(s)$,*

$$\frac{\mathbf{1}_{\{G_\lambda(s, \cdot) > 0\}} - \mathbf{1}_{\{G_\lambda(0, \cdot) > 0\}}}{s} = \int_0^1 \delta(G_\lambda(\theta s, \cdot)) \partial_s G_\lambda(\theta s, \cdot) d\theta \quad (96)$$

is a uniformly controlled coarea current. The omitted deeper words satisfy

$$|s|^{-1} C e^{-(c_{\text{tail}} - A_{\text{loss}})L(s)} \leq C|s|^{\kappa(c_{\text{tail}} - A_{\text{loss}}) - 1} = o(1). \quad (97)$$

Consequently the collision and grazing cutoffs can be removed simultaneously with the parameter limit; no fixed neighborhood is asserted to be uniformly transverse at every depth.

Proof. Lemma 7.4 gives the coarea identity and all geometric bounds along θs for shallow words. The return/grazing tail, the joint mark- Z ledger, and the allowed geometric loss give $C e^{-(c_{\text{tail}} - A_{\text{loss}})L}$ before the difference quotient. Apply the total-variation bound to both endpoints for words deeper than $L(s)$ and divide by $|s|$. The first inequality in (95) gives (97). $\square$

This construction is why a differentiable parameter family of Young towers is unnecessary. The dynamical return alphabet is fixed on clean master cells, while every atom which changes its itinerary is represented by the physical coarea current that the response formula already contains.

The clean cohomological bracket and the master refinement can now be put on one ledger. The point of the next proposition is not to create a second symbolic dynamics. It records, on the amplitude side, which already existing full branches and recovery descendants contain a physical face piece.

Proposition 7.6 (Global physical ledger). *Assume the exponential return/context moments, $\text{SS}(d_Z)$, $\text{PR}(\alpha_L, d_Z)$, $\text{FZ}(\theta_Z)$, and the pathwise jet condition of Definition 7.3. Let $q_H, q_F, q_D > 1$ be the available context, flux, and dominant-scale moment exponents, and assume*

$$q_H^{-1} + q_F^{-1} + q_D^{-1} < 1. \quad (98)$$

After artificial-face cancellation, the complete first defect has a disjoint signed refinement

$$\lambda = (\chi, a, k, c^-, g, c^+; Q; j, k', r, \sigma), \quad (99)$$

where χ is a physical chart, a a return word, k the insertion time, c^-, g, c^+ the single-source gate data, Q a native gate ball, (j, k', r) the incoming/outgoing strips and connected-component index, and σ the chosen time orientation. The refinement does not promote amplitude cuts to new inverse branches. For some $a_H > 0$ below the context moment threshold,

$$\sum_{\lambda} M_{\lambda} e^{\epsilon(|c^-| + |c^+|)} (1 + H_{\lambda})^{a_H} \mathfrak{d}_{\lambda}^3 \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_{\lambda}^-) + Z(\mathcal{G}_{\lambda}^+)\} < \infty. \quad (100)$$

All constants in the functional-correlation, native-energy, fractional-phase, return, grazing, and direct-Z estimates are uniform on the quantitative table family.

Proof. Intersect a finite physical coarea chart with one original full return branch, the native gate cover, and the two homogeneity partitions. Boundaries have zero face mass after coincident germs are grouped, so the inherited label is unique almost everywhere. The context and H moments follow from the full-branch Gibbs quotient and exponential gate hitting. Proposition 6.42 supplies the component-indexed joint mark-boundary factor on the same labels, before generalized Hölder is applied. The dominant-scale moment supplies $\mathfrak{d}_{\lambda}^3$. Return/grazing tails and Proposition 7.5 sum the remaining labels; the strict margin (94) absorbs every finite-depth geometric loss. Generalized Hölder is used with the three exponents in (98); no unrecorded $p > 2$ moment is chosen. This proves (100); it is not assumed again in the main theorem. $\square$

Theorem 7.7 (Pathwise quantitative CM2 theorem). *Let an analytic-boundary or circular deformation $a(s)$ satisfy QT_{path} . Assume a compact finite-horizon chart with:*

- (H1) *the finite-horizon functional-correlation bound of [5], the standard one-state marked quotient, and exponential return/context moments; a dominant inverse-scale moment $\sum_{\lambda} M_{\lambda} \mathfrak{d}_{\lambda}^{3q_D} < \infty$; and exponents q_H, q_F, q_D satisfying (98);*
- (H2) *forward/reverse proper-family growth and coupling, together with the signed multiplier Lemma 6.45;*
- (H3) *$\text{SS}(d_Z)$ and $\text{PR}(\alpha_L, d_Z)$ on the persistent clean gate;*
- (H4) *the two-chart nondegeneracy $|\nabla G| \geq c_G$ and $\text{FZ}(\theta_Z)$, including its exact two-flux identity, weighted component complexity, angle sublevel, and good-angle exponent A_{ang} ;*
- (H5) *the physical event-current bound (69), with $1 \leq R \leq 3$ and observable regularity above the dominant two-dimensional Fourier threshold.*

Then the complete double-centered physical first defect satisfies

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}.$$

The assembled difference quotients converge after the depth-dependent split $L(s)$ of (95), and their CM2 series converges term by term as $s \rightarrow 0$. More precisely, if c_{off} is the off-diagonal coupling exponent and c_Z is the exponent obtained by balancing the FZ angle sublevel, one may choose

$$c_* < \min\{c_{\text{off}}, c_{\text{FC}} - A_{\text{dom}}\eta, \eta\beta_{\text{cen}}, \eta R, c_{\text{tail}} - A_{\text{loss}}, c_Z\} \quad (101)$$

and then take $\tau = \vartheta = e^{-c_*/2}$ after changing the constant.

Proof. Conditions SS and PR, together with the mass-preserving buffer, give Theorem 6.25. The high-frequency balanced estimate is Proposition 6.33. The regular two-dimensional defect is included in the low-frequency Proposition 6.37. Choosing $K = e^{\eta \max\{m, n\}}$ produces the low, face-high, and regular-tail exponents in (101).

Outside the diagonal band use Proposition 6.46; Lemma 6.2 converts the three estimates to product decay. For radial pieces Proposition 6.42, Lemma 6.43, and Corollary 6.44 give the common physical majorant without a component-length estimate and with every component explicitly summed. Proposition 7.6 gives the global physical ledger from the primitive hypotheses. Lemma 7.4 and Proposition 7.5 give coarea convergence up to $L(s)$ and an $o(1)$ quotient tail beyond $L(s)$. No fixed neighborhood is claimed to be transverse at every depth, and no qualitative residual assertion is used in this step. $\square$

Proposition 7.8 (Pathwise coarea-current convergence). *In a compact table chart satisfying (H1)–(H5), all constants in the functional-correlation, native-energy, fractional-phase, direct-Z, and parameter-tail estimates are uniform up to $L(s)$. The shallow coarea current has the CM2 majorant of Theorem 7.7, while the deep remainder is $o(1)$ in the CM2 sum.*

Proof. Hyperbolicity, distortion, the clean gate, and single-face nondegeneracy are uniform on the quantitative compact atlas. The functional-correlation constants are uniform for a compact finite-horizon family, and the return/context moments and both one-sided coupling estimates are uniform by (H1)–(H2). Proposition 6.42 fixes the component-indexed density/boundary majorant. Use the strict path margin (94). Proposition 7.5 supplies the depth-dependent scalar quotient tail. Apply the low/high central estimates and the off-diagonal couplings label by label to the shallow part, sum (100), and use (97) for the deep part. $\square$

The construction does not require a differentiable parameter family of countable towers. The coarea depth grows like $L(s)$, while the remaining collision/grazing words are removed by the quotient tail (97).

8 Global ledger and quantitative falsification checks

The proof has five independent exponent ledgers.

(A1) In the diagonal band the low-frequency ledger is

$$K^{A_{\text{dom}}\theta_{\text{FC}}^{\max\{m, n\}}}, \quad K = e^{\eta \max\{m, n\}}.$$

(A2) The high-frequency ledger is $K^{-\beta_{\text{cen}}}$, obtained from the native single-source energy and fractional phase.

- (A3) The regular high-frequency ledger is K^{-R} from Lemma 6.36.
- (A4) Lemma 6.2 supplies the two off-diagonal coupling ledgers.
- (A5) The physical majorant is the pathwise return/grazing quotient tail multiplied by the joint mark- Z sum (82).

Choosing η as in Proposition 6.37 makes all five exponents positive and gives (101). Zero and axis modes are removed by global centering before the physical chart sum; they are not estimated label by label. The regular two-dimensional defect is included in the same functional-correlation estimate.

There are four falsifiable geometric interfaces: SS must contain exactly one Leclerc rough source; PR must provide actual-SRB regeneration into legal native children; FZ must count every connected component, prove the exact two-flux Jacobian, and record A_{ang} ; and the chosen deformation must verify QT_{path} with the strict margin (94). These are primitive quantitative conditions, not consequences hidden in a Banach norm or a qualitative genericity statement.

References

- [1] M. Demers and C. Liverani, *Projective cones for sequential dispersing billiards*, Commun. Math. Phys. 401 (2023), arXiv:2104.06947.
- [2] M. Demers and C. Liverani, *Recent progress in the application of transfer operators to dispersing billiards*, arXiv:2606.10155 (2026).
- [3] M. Demers and C. Liverani, *Central limit theorem for sequential dynamical systems*, arXiv:2502.07765v2 (2025).
- [4] M. Stenlund, L.-S. Young and H.-K. Zhang, *Dispersing billiards with moving scatterers*, arXiv:1210.0011.
- [5] J. Leppänen and M. Stenlund, *A note on the finite-dimensional distributions of dispersing billiard processes*, arXiv:1702.01461.
- [6] N. Fleming-Vázquez, *Functional correlation bounds and optimal iterated moment bounds for slowly-mixing nonuniformly hyperbolic maps*, arXiv:2106.06486v2.
- [7] I. Baldomá, A. Florio, M. Leguil and T. M.-Seara, *Chaoticity of generic analytic convex billiards*, arXiv:2605.19897 (2026).
- [8] C. Wormell, *Conditional mixing in deterministic chaos*, arXiv:2206.09291; and *On convergence of linear response formulae in piecewise hyperbolic maps*, arXiv:2206.09292.
- [9] S. Baker and A. Banaji, *Polynomial Fourier decay for fractal measures and their pushforwards*, Math. Ann. 392 (2025), 209–261, arXiv:2401.01241.
- [10] S. Baker, O. Khalil and T. Sahlsten, *Fourier decay from L^2 -flattening*, arXiv:2407.16699.
- [11] V. Climenhaga and J. Day, *Every finite horizon Sinai billiard map has a unique measure of maximal entropy*, arXiv:2604.25881.

- [12] P. Bálint, P. Nándori, D. Szász and I. P. Tóth, *Equidistribution for standard pairs in planar dispersing billiard flows*, Ann. Henri Poincaré 19 (2018), arXiv:1707.03068.
- [13] O. Butterley, G. Canestrari and R. Castorrini, *Discontinuities cause essential spectrum on surfaces*, Ann. Henri Poincaré (2024), arXiv:2306.00484.
- [14] G. Canestrari, *Linear response for Sinai billiards with small holes*, arXiv:2604.19671v2 (2026).
- [15] G. Leclerc, *Fourier decay of equilibrium states for bunched attractors*, arXiv:2301.10623.
- [16] G. Leclerc, *Fourier decay of equilibrium states and the Fibonacci Hamiltonian*, arXiv:2507.23731v2 (2026).
- [17] G. Leclerc, S. Paukkonen and T. Sahlsten, *Fourier decay in parabolic $C^{1+\alpha}$ systems with overlaps*, arXiv:2505.15468v3 (2025).
- [18] A. Algom, F. Rodriguez Hertz and Z. Wang, *Polynomial Fourier decay and a cocycle version of Dolgopyat’s method for self conformal measures*, arXiv:2306.01275v2 (2024).
- [19] Y. Lima, D. Obata and M. Poletti, *Measures of maximal entropy for non-uniformly hyperbolic maps*, arXiv:2405.04676v2 (2025).
- [20] N. Chernov, *Decay of correlations and dispersing billiards*, J. Stat. Phys. 94 (1999), 513–556.
- [21] P. Teixeira, *On the conservative pasting lemma*, Ergodic Theory Dynam. Systems 40 (2020), 1402–1440, arXiv:1611.01694.
- [22] R. Rühr, *Pressure inequalities for Gibbs measures of countable Markov shifts*, arXiv:2012.13226v3.
- [23] N. Bertozzi, *Exponential mixing for hyperbolic flows on non-compact spaces*, arXiv:2603.11852v2 (2026).
- [24] V. Climenhaga, *Gibbs measures have local product structure*, arXiv:2310.17495v2 (2025).
- [25] S. Gouëzel, *Berry–Esseen theorem and local limit theorem for non uniformly expanding maps*, Ann. Inst. H. Poincaré Probab. Statist. 41 (2005), 997–1024, arXiv:math/0310391.
- [26] D. Galli, *A cohomological approach to Ruelle–Pollicott resonances and speed of mixing of Anosov diffeomorphisms*, arXiv:2405.17045.